# Critical Behavior of the Clean Three-Dimensional Lattice $\phi^4$ Model

Midsemester Progress Report

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## 1 Refined goals and background

I am using the three-dimensional lattice $\phi^4$ model to study how an ordered state develops as temperature decreases. Each site has a real-valued field, and neighboring fields favor alignment. With one scalar order parameter, sign-reversal symmetry, and short-range interactions, I expect a continuous transition to fall in the three-dimensional Ising universality class.

My goal is to estimate $T_c$ and the exponents $\nu$, $\beta$, and $\gamma$ from several lattice sizes. These describe the correlation length, order parameter, and susceptibility near the transition. The current results use a fully occupied lattice with fixed couplings. I also want to explore how random vacancies affect the transition and why quenched disorder can lead to a different universality class.

One physical example is the Ising antiferromagnet $FeF_2$, where the field can represent staggered magnetization, the difference between the magnetizations of the two sublattices [1]. In a binary liquid near demixing, it can represent the local composition relative to the critical composition. Experiments on water–2,6-lutidine show how these fluctuations produce critical Casimir forces between a colloid and a surface [2]. Calculating those forces would require adding surfaces and their boundary conditions to this bulk model.

Since the proposal, I have implemented Metropolis sampling and parallel tempering and completed the first exponent fits. I will use these clean results as a reference when I add vacancies. The simulation and analysis code are in the project repository [9].

## 2 Implemented methods

### 2.1 Hamiltonian and local Metropolis updates

The simulation uses a periodic simple cubic lattice with $N = L^3$ sites and a real field $\phi_i$ at each site. The Hamiltonian is

$$H = -J \sum_{\langle ij \rangle} \phi_i \phi_j + \sum_i \left( \frac{r}{2} \phi_i^2 + \frac{u}{4} \phi_i^4 \right), \qquad J = 1, \quad r = -1, \quad u = 1, \tag{1}$$

where each nearest-neighbor bond is counted once. I set $k_B = 1$, so a configuration has weight proportional to $\exp(-H/T)$.

One sweep visits each site and proposes $\phi_i' = \phi_i + \delta$, with $\delta$ drawn uniformly from a fixed symmetric interval. If $S_i$ is the sum of the six neighboring fields, the energy change is

$$\Delta H = -J(\phi_i' - \phi_i) S_i + \frac{r}{2}(\phi_i'^2 - \phi_i^2) + \frac{u}{4}(\phi_i'^4 - \phi_i^4). \tag{2}$$

The proposal is accepted with probability $\min[1, \exp(-\Delta H/T)]$ [3]. I use both random and ordered starting fields. The proposal width is adjusted during the initial sweeps and held fixed afterward. I checked the local energy change against full-energy calculations and analytic identities.

### 2.2 Parallel tempering

Near the transition, large regions fluctuate together, and local updates change these regions slowly. I use parallel tempering so configurations can visit higher temperatures, where ordered regions break up and reorganize, before returning near $T_c$ [4]. These visits can help reduce correlations between measurements. How well this works depends on the local Metropolis updates and how long replicas spend at the other temperatures; slow fluctuations can still remain.

Replicas run at different temperatures and attempt exchanges between neighboring pairs, alternating even and odd pairs. For total energies $H_a, H_b$ at $T_a < T_b$, the exchange probability is

$$P_{\text{swap}} = \min\left\{1, \exp\left[\left(\frac{1}{T_a} - \frac{1}{T_b}\right)(H_a - H_b)\right]\right\}. \tag{3}$$

I group measurements by temperature after exchanges. The main runs use $L = 16, 24, 32, 48$, with 17, 25, 25, and 40 temperatures spanning 14.9–16.5. For each size, I ran two independent temperature ladders: one from random fields and one from an ordered state, using separate random-number streams. Every temperature has 2,000 adaptation sweeps, 50,000 settling sweeps, and 500,000 measurement sweeps. Exchanges and saved measurements occur every five sweeps.

### 2.3 Observables and statistical errors

For each configuration, $e = H/N$ and $m = N^{-1}\sum_i \phi_i$. I calculate

$$B_4 = \frac{\langle m^4\rangle}{\langle m^2\rangle^2}, \qquad B_{\text{N}} = \frac{3 - B_4}{2}, \qquad \chi_{|m|} = \frac{N}{T}\left(\left\langle m^2\right\rangle - \langle|m|\rangle^2\right). \tag{4}$$

The order parameter is $\langle|m|\rangle$. The normalized Binder cumulant approaches one in the ordered limit and zero in the Gaussian limit.

I calculate autocorrelation separately for energy, $|m|$, $m^2$, and $m^4$ at each temperature and lattice size. For a history $x_1, \ldots, x_n$, I subtract its mean $\bar{x}$ and multiply pairs of deviations separated by $k$ saved measurements. Averaging these products gives

$$C_x(k) = \frac{1}{n-k}\sum_{i=1}^{n-k}(x_i - \bar{x})(x_{i+k} - \bar{x}), \qquad \rho_x(k) = \frac{C_x(k)}{C_x(0)}. \tag{5}$$

If measurements tend to stay on the same side of the mean, their products are positive. As this memory fades, $\rho_x(k)$ approaches zero. The normalization gives $\rho_x(0) = 1$. I estimate the integrated correlation time by adding the correlations over a range of time gaps:

$$\tau_{\text{int},x} = \frac{1}{2} + \sum_{k=1}^{W}\rho_x(k), \qquad n_{\text{eff}} \simeq \frac{n}{2\tau_{\text{int},x}}. \tag{6}$$

The upper limit $W$ keeps noisy estimates at large time gaps from dominating the sum. The correlation time is measured in saved-measurement units; each saved measurement is five sweeps apart. Here $n$ is the total number of saved measurements, and $n_{\text{eff}}$ is the estimated number of independent measurements that would give similar precision. Positive correlations reduce this effective count. This estimate assumes equilibrium and can miss weak, slow fluctuations. I also increase the block length to check whether the error estimates level off.

Point error bars come from leaving out one time block at a time and recalculating the observable (block jackknife). Central fits use 2,000 saved samples per block, and plotted error bars use 5,000.

## 3 Finite-size scaling and preliminary results

Near $T_c$, the correlation length becomes comparable to the lattice size. Observables at the transition then scale with $L$ [5]:

$$\left|\frac{dB_4}{dT}\right| \propto L^{1/\nu}, \qquad \langle|m|\rangle \propto L^{-\beta/\nu}, \qquad \chi_{|m|} \propto L^{\gamma/\nu}. \tag{7}$$

I fit quadratic temperature curves over $15.30 \le T \le 15.34$ and find the $L = 24, 48$ Binder crossing at $T_c \simeq 15.3101$. Weighted log–log fits at this temperature give powers $1/\nu = 1.640$, $-\beta/\nu = -0.519$, and $\gamma/\nu = 1.957$ (Figure 2).

For uncertainty, I resample time blocks with replacement and repeat the crossing and both fitting stages 2,000 times. Each ladder uses the same selected blocks across all temperatures and moments. The two independent ladders are resampled separately, then combined with their original sample weights. Point-error weights stay fixed. Table 1 gives the middle 95% of the resulting $T_c$ and exponent estimates.

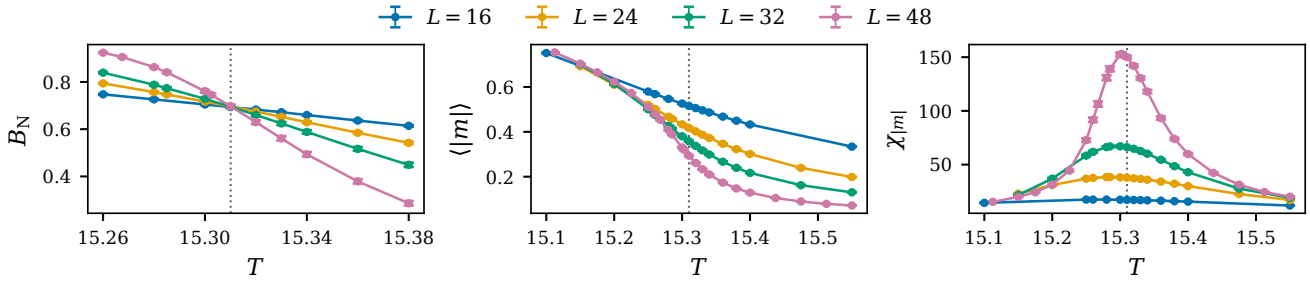



**Figure 1:** Binder crossing, magnetization, and susceptibility for $L = 16, 24, 32, 48$. Error bars are one block-jackknife standard error using 25,000-sweep blocks. The dotted line marks the fitted crossing temperature.

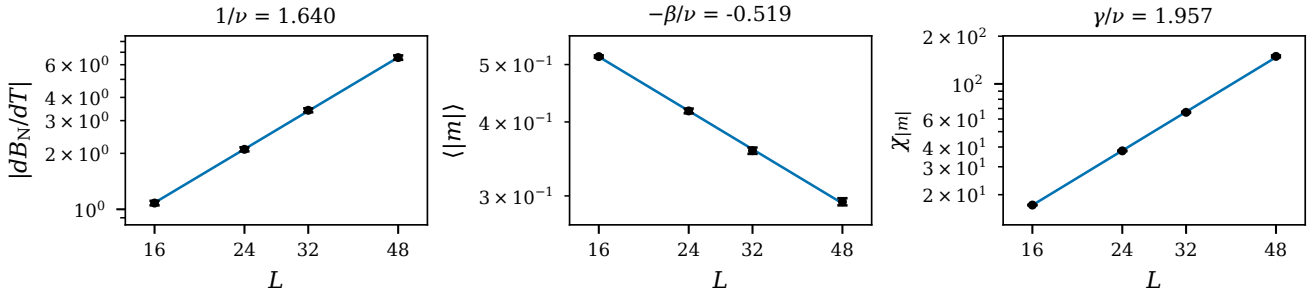



**Figure 2:** Size scaling at $T_c \simeq 15.3101$, on logarithmic axes. Points are interpolated values; lines are the weighted power-law fits. The slopes give $1/\nu$, $-\beta/\nu$, and $\gamma/\nu$. Error bars are one bootstrap standard error, including uncertainty in $T_c$.

**Table 1:** Effective finite-size estimates and marginal 95% statistical intervals. The Ising comparison uses Ref. [6]; $\beta$ and $\gamma$ are derived from its $\nu$ and $\eta$.

| Quantity | Estimate | 95% interval | 3D Ising comparison |
|---|---|---|---|
| $T_c$ | 15.3101 | [15.3069, 15.3137] | model dependent |
| $\nu$ | 0.610 | [0.580, 0.639] | 0.63002 |
| $\beta$ | 0.316 | [0.292, 0.342] | 0.32644 |
| $\gamma$ | 1.193 | [1.141, 1.243] | 1.23719 |

The susceptibility peak rises from 17.45 at $L = 16$ to 152.93 at $L = 48$. All three exponent intervals include the literature values; longer runs and larger lattices are needed to check the remaining shifts.

# 4 Difficulties and solutions

## 4.1 Slow fluctuations and finite histories

The first short runs had broad susceptibility peaks and strongly correlated measurements. The peak became clearer at larger sizes. Some replicas still move slowly through the temperature ladder: three in the $L = 48$ runs made no full round trip. Errors for some observables grow by 20–30% when block length increases from 1,000 to 5,000 saved samples. The longer choice leaves only 20 blocks per ladder, so I need longer histories to check where the errors level off.

## 4.2 How the fitting choices affect the exponents

For the main estimate, I fit temperatures from 15.30 to 15.34. Widening this to 15.28–15.36 shifts $\nu$ to about 0.654. Over the wider range, a quadratic fits the $L = 48$ susceptibility poorly: the residuals are large compared with the point errors. Figure 3 shows this result in red. I have kept it in the comparison because the shift is larger than those from the other fitting choices.

The other rows change the block length, use a linear temperature fit, remove $L = 16$, change the Binder crossing pair, or fit each initial-state run separately. These changes produce smaller shifts. Removing $L = 16$ checks the influence of small lattices, and fitting each start separately checks whether the two runs give consistent estimates. Longer runs and larger sizes should make these comparisons more informative.

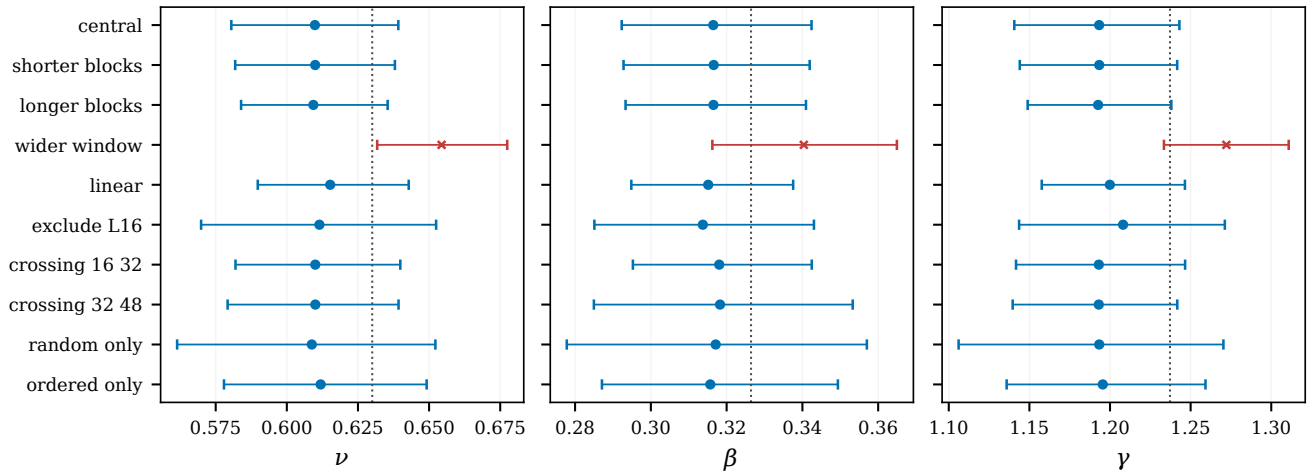



**Figure 3:** Each row changes one analysis choice. Bars are 95% statistical intervals; dotted lines are the Ising reference. The red wider-window result has a poor local quadratic fit.

**Hyperscaling as a consistency check.** For a scale-invariant critical magnetization distribution, both $\langle m^2 \rangle$ and $\langle |m| \rangle^2$ scale as $L^{-2\beta/\nu}$. The susceptibility definition then gives $\chi_{|m|} \propto L^{d-2\beta/\nu}$. Comparing this with $L^{\gamma/\nu}$ gives the hyperscaling relation

$$\frac{2\beta}{\nu} + \frac{\gamma}{\nu} = d = 3. \tag{8}$$

The fitted combination is 2.994, with a 95% interval of [2.954,3.038]. The powers are fitted independently before making this comparison.

## 5 Plan for completion and quenched disorder

**Finish the clean calculation.** I will extend the $L = 32, 48$ runs to check block-error stability and add a full temperature ladder at $L = 64$. Repeating the fits across windows and sets of sizes will help separate interpolation effects from finite-size corrections. Comparisons of independent starts and production halves will continue to check the sampling.

**Explore why disorder can change universality.** I would like to randomly remove a small fraction of sites, keep that vacancy pattern fixed throughout each simulation, and repeat the calculation for several patterns. This is quenched disorder: the fields fluctuate, while the vacancies stay fixed. A small comparison could show how vacancies change the ordering curves, susceptibility peak, and transition temperature. Differences between vacancy patterns must be averaged as well as the thermal fluctuations within each pattern.

For weak, uncorrelated dilution, the Harris criterion predicts that disorder changes the clean critical behavior when $d\nu_{\text{clean}} < 2$ [7]. Here $3(0.63002) \simeq 1.89 < 2$, so disorder is a relevant perturbation. Published simulations find a randomly diluted Ising universality class with $\nu \simeq 0.683$, compared with the clean value near 0.630 [8].

For this stage, I want to show how vacancies affect the transition and look for changes in the size dependence. I do not expect enough time to measure the disordered exponents precisely. A shift in $T_c$ alone would not establish new exponents. I will compare what the runs show with the Harris criterion and published results, and be clear about which scaling differences the data can resolve.

## References

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